

A Finite-versus-Infinite Contradiction in Conjecture 00000008304

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The incompatible clauses

The conjecture calls `brk_pts` the breakpoint set of the packing function and asserts that this set is finite with exact count $2^n - n - 1$. It also states that the threshold for infinitely many breakpoints is 24, specifically saying that there are infinitely many breakpoints for $n \geq 24$. No different breakpoint set or parameter is introduced between these clauses.

Contradiction at $n = 24$

The exact-count assertion makes the breakpoint set at $n = 24$ equivalent to a finite set with $2^{24} - 24 - 1 = 16777191$ elements. In particular, it is finite. The threshold assertion says that the same set at $n = 24$ is infinite. A set cannot be both finite and infinite. Thus these two clauses cannot simultaneously hold, so the full conjecture, which includes both, is false as written.

Formalization boundary

Lean represents the breakpoint sets by an arbitrary family of types indexed by n . It expresses finite cardinality as equivalence to a standard finite type and infinite cardinality as the negation of that property. The contradiction is proved at $n = 24$ without formalizing symplectic ball packings, EHZ capacities, or E8 lattices, because those other conjecture clauses cannot repair the inconsistent pair.